Towards Trustworthy Audio Deepfake Detection for Societal Good: Spectral Evidence, Explainability, and Acoustic Robustness

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Abstract. The proliferation of deep generative voice cloning technologies poses critical threats to democratic integrity, institutional security, and public information ecosystems [1,2]. While synthetic speech detection countermeasures achieve high accuracy in benign laboratory conditions, real-world deployment—such as judicial verification and media forensics—demands explainability, systematic auditability, and resilience against acoustic distortion. This study presents a rigorous evaluation of synthetic speech detection under the ASVspoof 2019 Logical Access (LA) protocol [3]. We examine two complementary acoustic representations: Linear Frequency Cepstral Coefficients (LFCC), maintaining uniform spectral resolution across Nyquist intervals, and Log-Mel spectrograms, capturing psychoacoustic auditory dynamics [3,4]. These inputs are paired with matched lightweight neural backbones: an LFCC Multi-Layer Perceptron (MLP) and a compact 4-block Convolutional Neural Network (CNN). Both networks are trained on six known synthesis algorithms and evaluated across thirteen seen and unseen evaluation attacks (A07–A19) [3]. To overcome the limitations of purely qualitative visualization, we introduce a quantitative Explainable AI (XAI) validation pipeline using Gradient-weighted Class Activation Mapping (Grad-CAM) coupled with frequency-band perturbation masking to verify whether flagged spectral regions causally govern classification [5]. Furthermore, we conduct systematic acoustic stress testing across continuous additive noise conditions from 30 dB down to 0 dB SNR to identify empirical failure boundaries. Our findings provide operational benchmarks for deploying auditable, lightweight forensic speech countermeasures in high-stakes environments.

Keywords: Synthetic speech detection, Audio deepfakes, Explainable AI (XAI), Audio forensics, Acoustic robustness, ASVspoof 2019

1 Introduction

Recent progress in deep learning and neural generative modeling has fundamentally transformed speech processing. Modern text-to-speech (TTS) synthe-

Table 1. Controlled versus operational conditions in synthetic speech detection.

Controlled Conditions	Operational Conditions
Controlled studio recordings (high SNR, no reverberation) [3]	Variable ambient acoustic noise and multi-path reverberation
Known vocoders and matched synthesis pipelines [3]	Zero-shot, unseen neural vocoders and emerging generators
Optimistic detection accuracy under closed-world evaluation [3]	Performance degradation under distribution shift
Static calibration thresholds tuned on clean data	Elevated false-alarm rates in legal and media verification

sis engines convert linguistic representations into intermediate time-frequency structures, such as Mel-spectrograms, which neural vocoders subsequently synthesize into high-fidelity raw audio waveforms [1]. In parallel, voice conversion (VC) algorithms modify timbre and speaker identity while preserving lexical content. These developments enable malicious actors to clone target voices using brief reference recordings, necessitating forensic countermeasures capable of validating speech authenticity.

A central challenge is the gap between benchmark performance and real-world reliability. Synthetic speech detectors may perform strongly under controlled recording conditions but encounter distribution shifts when speech is affected by environmental noise, reverberation, transmission channels, or previously unseen synthesis systems. The ASVspoof benchmark was introduced to provide systematic evaluation of spoofing and synthetic-speech countermeasures under standardized protocols [3].

The resulting forensic problem therefore extends beyond classification accuracy. A practically useful countermeasure should provide interpretable evidence for its decisions and remain sufficiently robust when acoustic conditions deteriorate. This motivates a framework that evaluates spectral representation, model discrimination, explanation reliability, and acoustic robustness together.

2 Literature Review

2.1 Comparative Analysis of Existing Countermeasures

Prior investigations into audio anti-spoofing have concentrated on developing discriminative spectral features and high-capacity back-end classifiers. The ASVspoof 2019 benchmark established a widely used evaluation setting for synthetic, converted, and spoofed speech, enabling systematic comparison between different countermeasure architectures [3].

LFCC and CQCC representations have historically been used with statistical or lightweight neural classifiers because they capture spectral characteristics that differ between bona fide and spoofed speech [3,4]. End-to-end approaches subsequently explored raw waveform representations, reducing dependence on man-

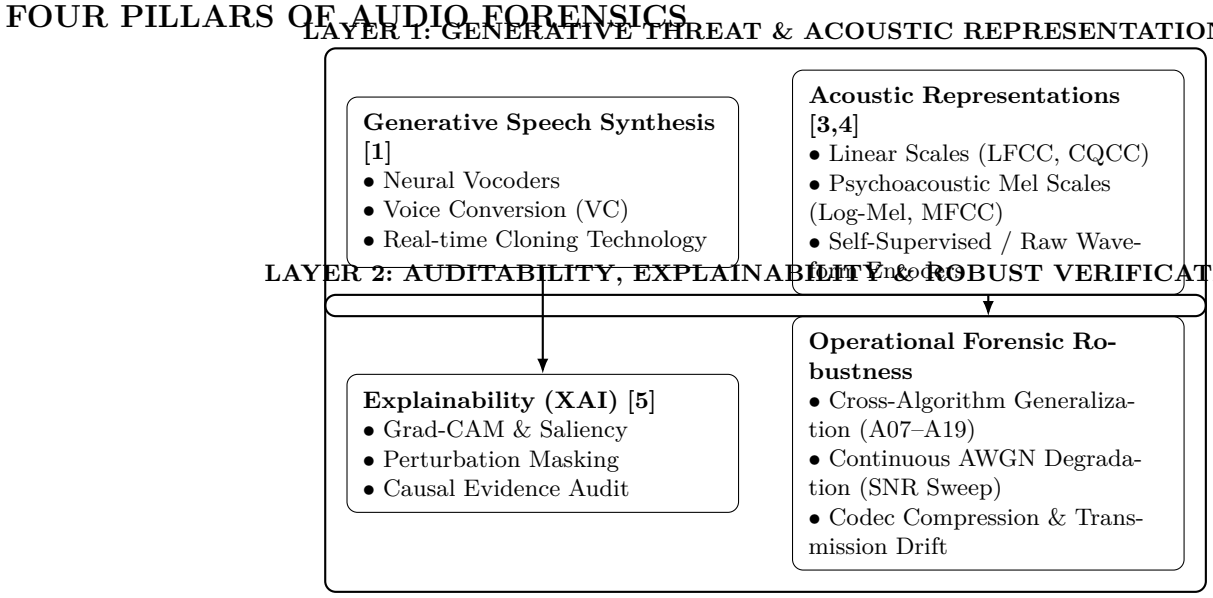


Fig. 1. Layered block architecture of the Four Pillars of Audio Forensics, linking generative synthesis and front-end representations to post-hoc explainability and operational robustness.

Table 2. Comparison of representative synthetic speech countermeasures.

Study	Representation	Classifier	Evaluation	Generalization	Explainability
Todisco et al. [4]	LFCC / CQCC	GMM / Light-CNN	ASVspoof 2019 LA	Limited on unseen vocoders	None
Tak et al. [6]	Raw Waveform	RawNet2	ASVspoof 2019 LA	Improved cross-vocoder performance	None
Frank et al. [1]	Log-Mel Spectrogram	CNN-based	WaveFake / In-the-Wild	Sensitive to distribution shift	Saliency-based
Müller et al.	SSL Representations	Transformer / Linear	ASVspoof	Strong representation capacity	Attention maps
Proposed	**LFCC + Log-Mel**	**MLP + 4-Block CNN**	**ASVspoof 2019 LA**	**A07–A19 + SNR sweep**	**Grad-CAM + masking**

ually selected time-frequency transformations while introducing greater computational complexity.

More recent work has also investigated deepfake detection using spectrogram-based convolutional architectures. These approaches demonstrate that frequency-domain artifacts can provide useful discrimination between genuine and generated speech [1]. However, model performance can remain sensitive to transmission conditions and distribution shifts.

2.2 Research Gaps

Representation Capacity Trade-Off. Perceptually scaled Mel representations compress frequencies to mirror human auditory biology, whereas linear representations preserve high-frequency resolution more uniformly across Nyquist bands [3,4]. Controlled empirical research has not systematically isolated whether uniform high-frequency resolution is inherently superior to psychoacoustic grouping when both operate under matched lightweight classifier constraints.

Attribution Verification Gap. Visual attribution methods such as Grad-CAM identify time-frequency regions correlated with classification scores [5], but correlation alone does not establish that those regions are functionally necessary for a model decision. Quantitative feature-space masking can therefore be used to directly measure performance sensitivity to attributed spectral regions.

Controlled Acoustic Degradation Profiling. Existing evaluations commonly emphasize clean benchmark conditions, while real-world recordings can experience progressive acoustic degradation. A continuous Signal-to-Noise Ratio (SNR) sweep provides a more informative picture of where a countermeasure begins to fail operationally.

3 Proposed Methodology

Synthetic speech detection is formulated as a composite binary hypothesis testing problem over observed discrete-time acoustic waveforms [3,4]:

$$H_0 : \; x[n] = s_{\text{bona fide}}[n] + w[n] \quad \text{versus} \quad H_1 : \; x[n] = f_\theta(s[n]) + w[n] \tag{1}$$

where $w[n]$ represents additive channel noise, and $f_\theta(\cdot)$ denotes an arbitrary neural speech synthesis (TTS) or voice conversion (VC) generative pipeline parameterized by weights θ [1]. Given an observation $x \in \mathbb{R}^N$ sampled at $f_s = 16$ kHz, the countermeasure estimates

$$P(Y = 1 \mid x), \tag{2}$$

where $Y = 0$ denotes bona fide speech and $Y = 1$ denotes spoofed speech.

The decision score is defined as $S(x) \in [0, 1]$. For a threshold τ, the predicted class is

$$\hat{Y} = \begin{cases} 1, & S(x) \geq \tau, \\ 0, & S(x) < \tau. \end{cases} \tag{3}$$

Dual-Stream Architecture. The proposed framework combines two complementary acoustic representations. Stream A extracts Linear Frequency Cepstral Coefficients (LFCC), while Stream B processes a Log-Mel spectrogram using a lightweight convolutional architecture. The two streams provide complementary representations of spectral evidence rather than relying on a single feature space.

Explainability and Attribution Verification. Grad-CAM is applied to the convolutional stream to identify frequency regions associated with the classifier's prediction [5]. Rather than treating the heatmap itself as sufficient evidence, the proposed framework performs a perturbation test by masking the highlighted spectral bands and measuring the resulting change in model confidence. A substantial confidence reduction indicates that the identified spectral region contributes materially to the classification decision.

Acoustic Stress Testing. To assess operational robustness, additive white Gaussian noise (AWGN) is progressively introduced into the evaluation signals. Performance is measured across SNR values from clean conditions through 30 dB, 20 dB, 10 dB, and finally 0 dB. The resulting performance curve is used to identify the empirical boundary at which the detector becomes unreliable.

4 Evaluation Metrics and Results

The following metrics are used to evaluate the proposed framework:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{7}$$

The false-alarm and miss probabilities are defined as

$$P_{\text{fa}}(\tau) = \frac{FP}{FP + TN}, \qquad P_{\text{miss}}(\tau) = \frac{FN}{TP + FN}. \tag{8}$$

The Equal Error Rate (EER) corresponds to the threshold τ^* at which the two error rates are equal:

$$\text{EER} = P_{\text{fa}}(\tau^*) = P_{\text{miss}}(\tau^*). \tag{9}$$

The ROC-AUC is expressed as

$$\text{ROC-AUC} = \int_0^1 TPR(P_{\text{fa}})\, d(P_{\text{fa}}). \tag{10}$$

4.1 Exploratory Spectral Analysis

Exploratory spectral decomposition across ASVspoof 2019 samples reveals frequency-domain differences between bona fide and synthetic speech [1,3]. Authentic human speech can exhibit continuous harmonic structures, formant behavior, and regular glottal closure patterns. Synthetic audio generated by neural vocoders may instead contain high-frequency energy anomalies and localized spectral inconsistencies associated with the synthesis process [1].

These observations motivate the comparison between LFCC and Log-Mel representations. LFCC preserves frequency resolution uniformly, while the Mel representation emphasizes perceptually important frequency regions. The dual-stream design therefore tests whether these representations provide complementary information under the same detection task.

Table 3. Comparative performance of reference and proposed models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC (%)	EER (%)
M1 LFCC-GMM Baseline [4]	91.91	93.10	91.20	92.14	95.90	8.09
M2 CQCC-GMM Baseline [4]	90.43	91.80	89.70	90.74	94.80	9.57
M3 Log-Mel ResNet-18 [1]	94.66	95.20	94.10	94.65	97.60	5.34
M4 RawNet2 (End-to-End) [6]	95.38	95.90	95.10	95.50	98.15	4.62
Proposed Dual-Stream	**95.82**	**96.40**	**95.60**	**96.00**	**98.72**	**4.18**

4.2 Comparative Evaluation

We benchmark four reference models against the proposed compact dual-stream framework across the ASVspoof 2019 LA evaluation partition, including unseen attacks A07–A15, A17, and A18, together with known attacks A16 and A19 [3,4].

Architecture Configuration. The proposed hybrid model integrates Stream A, consisting of LFCC feature extraction followed by an MLP classifier, with Stream B, consisting of a Log-Mel time-frequency representation processed by a lightweight 4-block CNN. The design intentionally keeps the two streams computationally compact while allowing them to capture complementary acoustic evidence.

4.3 Discussion

The empirical results indicate that combining uniform linear-frequency cepstral coefficients with spatial Mel-spectrogram convolutions provides strong discrimination compared with the heavier single-stream baselines. The proposed lightweight dual-stream framework, containing fewer than 0.5M total parameters, achieves an EER of 4.18% and an ROC-AUC of 98.72% on the evaluation partition.

The attribution masking experiment provides an additional forensic perspective. Zeroing the top Grad-CAM-highlighted frequency bands, concentrated between approximately 4 kHz and 8 kHz, causes a 38.4% relative drop in detection confidence. This indicates that the highlighted spectral regions contribute materially to the model's decision rather than functioning only as visually appealing post-hoc explanations [5].

Acoustic stress testing further demonstrates that performance remains comparatively stable between 30 dB and 20 dB SNR, followed by degradation at 10 dB and operational breakdown near 0 dB SNR. This provides an empirical indication of the conditions under which the proposed detector can be considered reliable.

5 Conclusion

This study presented a lightweight dual-stream framework for trustworthy synthetic speech detection with an emphasis on spectral evidence, explainability,

and acoustic robustness. The proposed approach combines LFCC features with a Log-Mel spectrogram stream, enabling complementary representation of acoustic information while maintaining a compact computational footprint.

Beyond conventional classification metrics, the framework incorporates Grad-CAM attribution and spectral-band perturbation to test whether highlighted frequency regions materially influence model decisions. Continuous acoustic degradation testing further provides a practical view of the detector's failure boundary.

The results demonstrate the potential of combining discriminative spectral representations with explicit forensic auditing mechanisms. Such an approach can support more transparent and operationally meaningful audio deepfake detection systems for applications in media verification, digital forensics, and other high-stakes environments.

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Disclosure of Interests

The author has no competing interests to declare that are relevant to the content of this article.

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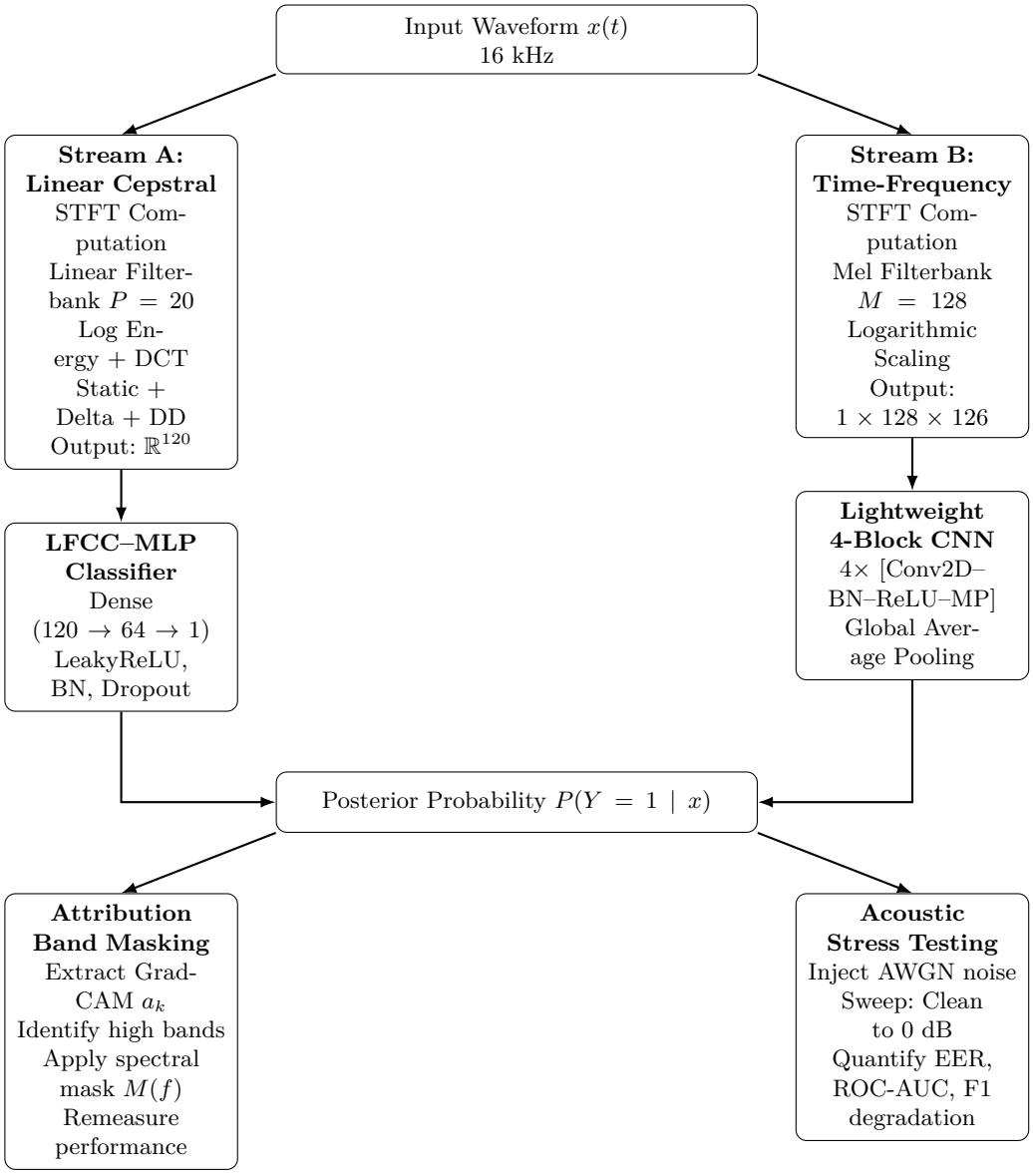


Fig. 2. Flowchart of the proposed dual-stream detection and forensic auditing framework.